Thm. Let F be a field, and put $V = P_n(F)$

$$\overset{def}{=} \left\{ \sum_{i=0}^{n} a_i x^i \,\middle|\, a_i \in F \right\}.$$

Suppose that $c_0, \cdots, c_n$ are distinct scalars. That is, $c_i \neq c_j$ $\forall i \neq j$.

Define

$$f_i : V \to F ;\ p(x) \mapsto p(c_i).$$

Then, $\{f_0, \cdots, f_n\}$ is dual basis of V^*.

Proof. Since $\dim V = \dim V^* = n+1$, enough to show that $\{f_0, \cdots, f_n\}$ is lin. independ..

$a_0 f_0 + \cdots + a_n f_n = 0$ Then, for $p(x) = (x - c_1) \cdots (x - c_n)$,

Let $f : F[x] \to F$, linear

$\exists p(x) \in F[x]$ s.t $f = \psi(p(x))$

Assume $V = F[x]$. Then,
$\beta = \{x^n \mid n = 0, 1, \cdots\}$ is basis of V, and
for $j = 0, 1, \cdots$, define

$$f_j : F[x] \to F : \sum_{i=0}^{n} a_i x^i \mapsto a_j$$

Then, $\beta^* = \{f_i \mid i = 0, 1, \cdots\}$ is dual basis of V^*.

And, $\phi : F[x] \to F[x]^* : \sum_{i=0}^{n} a_i x^i \mapsto \sum_{i=0}^{n} a_i f_i$

is injective linear map.

But, Cardinality of $F[x]^*$ is bigger than $F[x]$, because:
Let $f \in F[x]^*$ be given. That is, $f : F[x] \to F$ is a linear map.
f is determined by for each $i = 0, 1, \cdots$, $f(x^i) = a_i$.
Thus, $\#(F[x]^*) =$

「Thm. Let F be a field.
A Dual space of $F[x]$ is isomorphic to $F^{\mathbb{N}}$.

「proof. Consider a set of sequences:

$$F^{\mathbb{N}} := \{\{x_n\} : \mathbb{N} \to F\}.$$

with operation

$$\{x_n\} + \{y_n\} \stackrel{\text{def}}{=} \{x_n + y_n\}, \quad \alpha\{x_n\} \stackrel{\text{def}}{=} \{\alpha x_n\}.$$

Then, $(F^{\mathbb{N}}, +, \cdot)$ is Vector Space over F.

Define a map:

$$\phi : F[x]^* \to F^{\mathbb{N}} ; f \mapsto \{f(x^n)\}_{n=0}^{\infty}$$

linear: for $\alpha \in F$, $f, g \in F[x]^*$,

$f(x^n) = x_n$

$$\phi(\alpha f + g) = \{\alpha f(x^n) + g(x^n)\}$$
$$= \alpha\{f(x^n)\} + \{g(x^n)\} = \alpha\phi(f) + \phi(g).$$

Let $\phi(f)$ is zero. Then, for all $n = 0, 1, \cdots$, $f(x^n) = 0$.

This implies that f is zero map. $\therefore \phi$ is injective.

Let $\{x_n\} \in F^{\mathbb{N}}$ be given. Then, the

$$f : F[x] \to F : \sum_{i=0}^{n} a_i x^i \mapsto x_n \cdot a_n \cdot x^n$$

$$f\left(\sum_{i=0}^{n} a_i x^i + \sum_{i=0}^{n} b_i x^i\right)$$
$$= f\left(\sum_{i=0}^{n} (a_i + b_i) x^i\right)$$
$$= x_n (a_n + b_n) x^n$$
$$= x_n a_n x^n + x_n b_n x^n = f\left(\sum_{i=0}^{n} a_i x^i\right) + f\left(\sum_{i=0}^{n} b_i x^i\right)$$

$$f\left(\alpha \cdot \left(\sum_{i=0}^{n} a_i x^i\right)\right)$$
$$= f\left(\sum_{i=0}^{n} \alpha a_i x^i\right) = x_n \alpha a_n x^n = \alpha \cdot x_n a_n x^n.$$

$\therefore f$ linear s.t. $\forall i = 0, 1, \cdots$, $f(x^i) = x_i$.

Thus, ϕ is onto.

Thm. Let V be an infinite dimensional Vector Space with a basis β.
Denote $\beta = \{v_i \mid i \in I\}$. And, for each $i \in I$,

$$f_i^* : V \to F : x \mapsto (\pi_i \circ \phi_\beta)(x)$$

Then, a set $\beta^* = \{f_i^* \mid i \in I\}$ can not spans V^*.

Proof. Define a map

$$\varphi : V^* \to F^{\mathrm{Card}(I)} ; f \mapsto (f(v_i))_{i \in I}.$$

$(\cdot)_{i \in I} : I \to F.$

$f : V \to F.$

$F[x].$

$F[x]$

Then, φ is isomorphism;
linear and injective are clear. And, to show that surjective;
Let $(a_i) : I \to F$ be given. Then, there exists a linear map

$$f : V \to F \text{ s.t } \forall i \in I,\ f(v_i) = a_i.$$

Thus, surjectivity is proved. Now, $V^* \cong F^{\mathrm{Card}(I)}$

$$\begin{tikzcd} V \arrow[r] \arrow[d, "\cong"'] & V^* \arrow[d, "\cong"] \\ F^{(I)} \arrow[r] & F^I. \end{tikzcd}$$

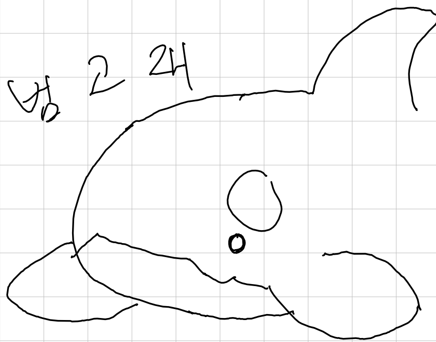
$$F^{(I)} := \{(x_i)_{i \in I} \mid \{i \in I \mid x_i \neq 0\} \text{ is finite}\}$$

Then, $V \cong F^{(I)}$ because;

$$\varphi : V \to F^{(I)} ; x \mapsto [x]_\beta$$

is isomorphism.

범 고 래



Def. Let $T: V \to V$ be a linear operator on finite vector space V.

T is said to be Diagonalizable if:

for some ordered basis β, $[T]_\beta$ is diagonal matrix.

For non-zero vector $v \in V$, Suppose that there exists a $\lambda \in F$ s.t

$$T(v) = \lambda v$$

Then, λ is called eigenvalue,

v is called eigenvector.

Thm. **Let V be a finite vector space, and $T: V \to V$ be a linear operator.**

T is diagonalizable if and only if There exists a basis β consisting eigenvectors of T.

Proof. Suppose that T is diagonalizable. Put β be a basis s.t

$[T]_\beta$ is diagonal Matrix.

Denote $\beta = \{v_1, \cdots, v_n\}$. Then, for each $1 \le i \le n$,

$$T(v_i) = \lambda_i v_i \quad \text{for some } \lambda_i \in F.$$

Each v_i is not zero, because β is a basis.

Conversely, Suppose that $\beta = \{v_1, \cdots, v_n\}$

「Corollary. $A \in M_{n\times n}(F)$ is diagonalizable if and only if
There exists a basis β of F^n consisting eigenvectors of L_A.

Note: put $\beta = \{v_1, \cdots, v_n\}$ be a basis of F^n consisting eigenvectors of A. Then, for a Matrix

$$Q = \begin{pmatrix} v_1 & v_2 & \cdots & v_n \end{pmatrix}$$

$$D = Q^{-1}AQ \quad \text{where } D = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$$

$$AQ = \begin{pmatrix} Av_1 & Av_2 & \cdots & Av_n \end{pmatrix}$$

$$= \begin{pmatrix} \lambda_1 v_1 & \lambda_2 v_2 & \cdots & \lambda_n v_n \end{pmatrix}$$

$$QD = Q\left(\begin{pmatrix} \lambda_1 & 0 & 0 \\ & 0 & \\ 0 & & 0 \end{pmatrix} + \begin{pmatrix} 0 & & \\ & \lambda_2 & 0 \\ 0 & & 0 \end{pmatrix} + \cdots + \begin{pmatrix} 0 & & \\ & 0 & 0 \\ 0 & & \lambda_n \end{pmatrix}\right)$$

$$= \begin{pmatrix} \lambda_1 v_1 & \cdots & \lambda_n v_n \end{pmatrix}$$

Def. Let $A \in M_{n\times n}(F)$ be a matrix.
Define characteristic polynomial of A;

$$f : F \mapsto F ; t \mapsto \det(A - tI_n).$$

Lemma. Let $A \in M_{n\times n}(F)$, $\lambda \in F$ be a eigenvalue of A. Then,

$v \in F^n$ is eigenvector corresponding to λ iff $v \neq 0$, $(A - \lambda I_n)v = 0$.

Lemma. Let $A \in M_{n\times n}(F)$.

$\lambda \in F$ is eigenvalue of A iff $\det(A - \lambda I) = 0$.

Proof. λ is eigenvalue

$\Leftrightarrow \exists v \in F^n$ with $v \neq 0$ s.t. $Av = \lambda v$.

$\Leftrightarrow$ 〃 s.t. $(A - \lambda I)v = 0$.

$\Leftrightarrow A - \lambda I$ is not invertible.

$Ax = 0$ has non-zero solution

$\Leftrightarrow \ker L_A \neq \{0\}$

$\Leftrightarrow L_A$ is injective

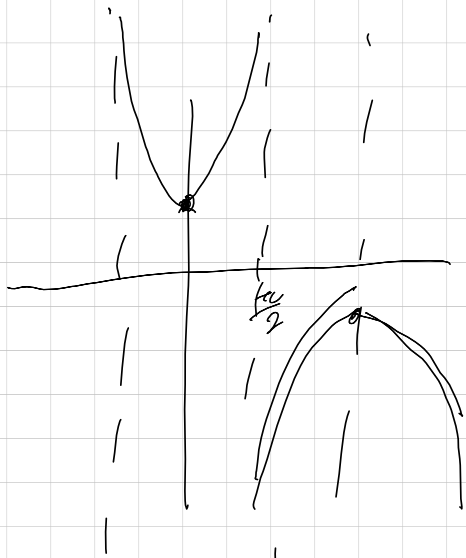
Thm. Let V be a finite vector space, $T: V \to V$ be a linear operator.
If β, γ are distinct ordered basis of V,
then $\det([T]_\beta) = \det([T]_\gamma)$.

Proof. Since

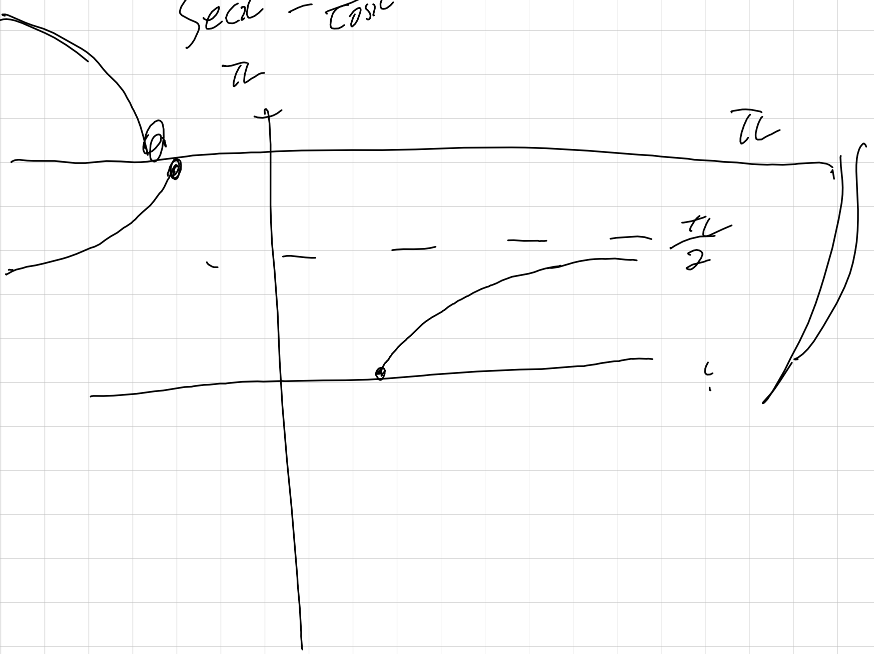
$$\det([T]_\beta) = \underline{\det([T]_\beta^\gamma)} \cdot \det([T]_\beta^\beta) \cdot \underline{\det([T]_\gamma^\beta)}$$

inverse

$$= \det([T]_\beta^\gamma [T]_\beta^\beta [T]_\gamma^\beta) = \det([T]_\gamma^\gamma)$$



$\sec x = \frac{1}{\cos x}$



「 f, g가 실수 전체에서 미분가능한 함수라 하자. 그럼 임의의 실수 a에 대해,

$$(f(g(a)))' = f'(g(a))\cdot g'(a)$$ 이다.

Proof.

Recall: $f'(x) = \lim_{t\to x} \frac{f(t)-f(x)}{t-x}$, if limit exists.

보이고 싶은 것: $\forall \varepsilon>0, \exists \delta>0$ s.t

$$0<|x-a|<\delta \Rightarrow \left| \frac{f(g(x))-f(g(a))}{x-a} - f'(g(a))\cdot g'(a)\right| < \varepsilon \left(\Leftrightarrow \lim_{t\to a} \frac{f(g(t))-f(g(a))}{t-a} = f'(g(a))\cdot g'(a) \right)$$

f, g가 각각 미분 가능하므로, 다음을 정의한다.

$$u(t) = \frac{g(t)-g(x)}{t-x} - g'(x) \text{ if } t\neq x, \quad u(t)=0 \text{ if } t=x$$

$$v(s) = \frac{f(s)-f(y)}{s-y} - f'(y) \text{ if } s\neq y, \quad v(s)=0 \text{ if } s=y.$$

로 정의한다. 그럼,

이제 임의의 $\varepsilon>0$이 주어졌다 하자.

$$\begin{cases} g(t)-g(x) = (t-x)\cdot(g'(x)+u(t)) \\ f(s)-f(y) = (s-y)\cdot(f'(y)+v(s)) \end{cases}$$ 을 만족한다.

이제, $$f(g(t))-f(g(x)) = (g(t)-g(x))\cdot(f'(g(x))+v(g(t)))$$
$$= (t-x)\cdot(g'(x)+u(t))\cdot(f'(g(x))+v(g(t)))$$ 이고, 즉

$$\frac{f(g(t))-f(g(x))}{t-x} = (g'(x)+u(t))\cdot(f'(g(x))+v(g(t)))$$ 이다.

$\lim_{t\to x} u(t) = 0$ 은 정의에 따라 자명하다.

$\lim_{t\to x} v(g(t)) = 0$ 을 보이자.

임의의 $\varepsilon>0$에 대해, $\delta>0$이 존재하여 $0<|y-\alpha|<\delta \Rightarrow \left|\frac{f(y)-f(\alpha)}{y-\alpha} - f'(y)\right| < \varepsilon$ 이다.

($\because$ f가 미분 가능)

이제, 이 $\delta>0$에 대해, $\delta'>0$이 존재하여 $|x-t|<\delta' \Rightarrow |g(x)-g(t)|<\delta$ 이다.

($\because$ g가 연속)

즉, $\forall \varepsilon>0\ \exists \delta'>0$ s.t $|x-t|<\delta' \Rightarrow |g(x)-g(t)|<\delta \Rightarrow \left|\frac{f(g(x))-f(g(t))}{g(x)-g(t)} - f'(g(x))\right| < \varepsilon$

다시 말해, $\lim_{t\to x} v(g(t)) = 0$. 이제, 아래의 양 변에 $\lim_{t\to x}$를 취하면 증명이 끝난다.

$$\frac{f(g(t))-f(g(x))}{t-x} = (g'(x)+u(t))\cdot(f'(g(x))+v(g(t)))$$

$[x, iy]$

$[x+y, i(x+y)]$

$= [x, ix] + [y, ix] + [x, iy] + [y, iy]$

$= 0 + \quad + \quad + 0$

$b[ix, y]$

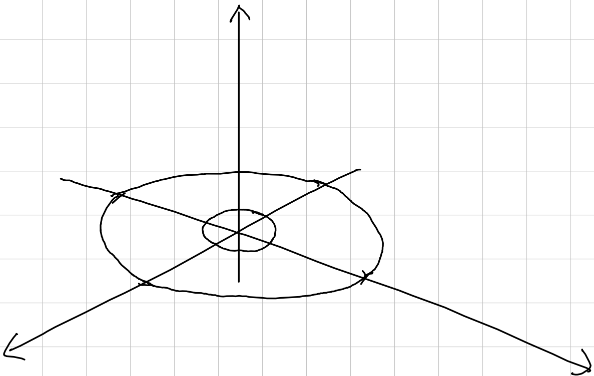
$= -b[x, iy]$

$= +i$

$\overline{\langle x, y \rangle}$

$= [x, y] - i[x, ix]$

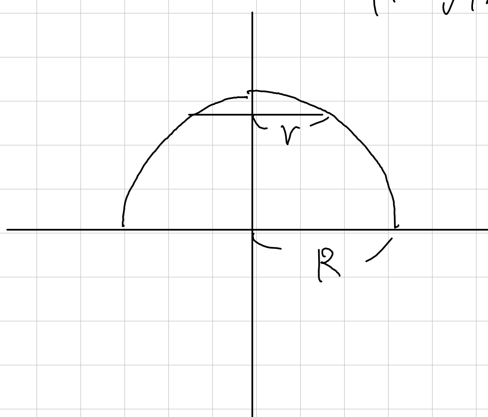
$= [\cancel{y}, i$



$$x = \sqrt{r^2 - y^2}$$

$$A(y) = (R^2 - y^2)\pi - r^2\pi$$

$$R - \sqrt{R^2 - r^2}$$



$$V = \int_a^b 2\pi x \cdot f(x) \cdot dx.$$